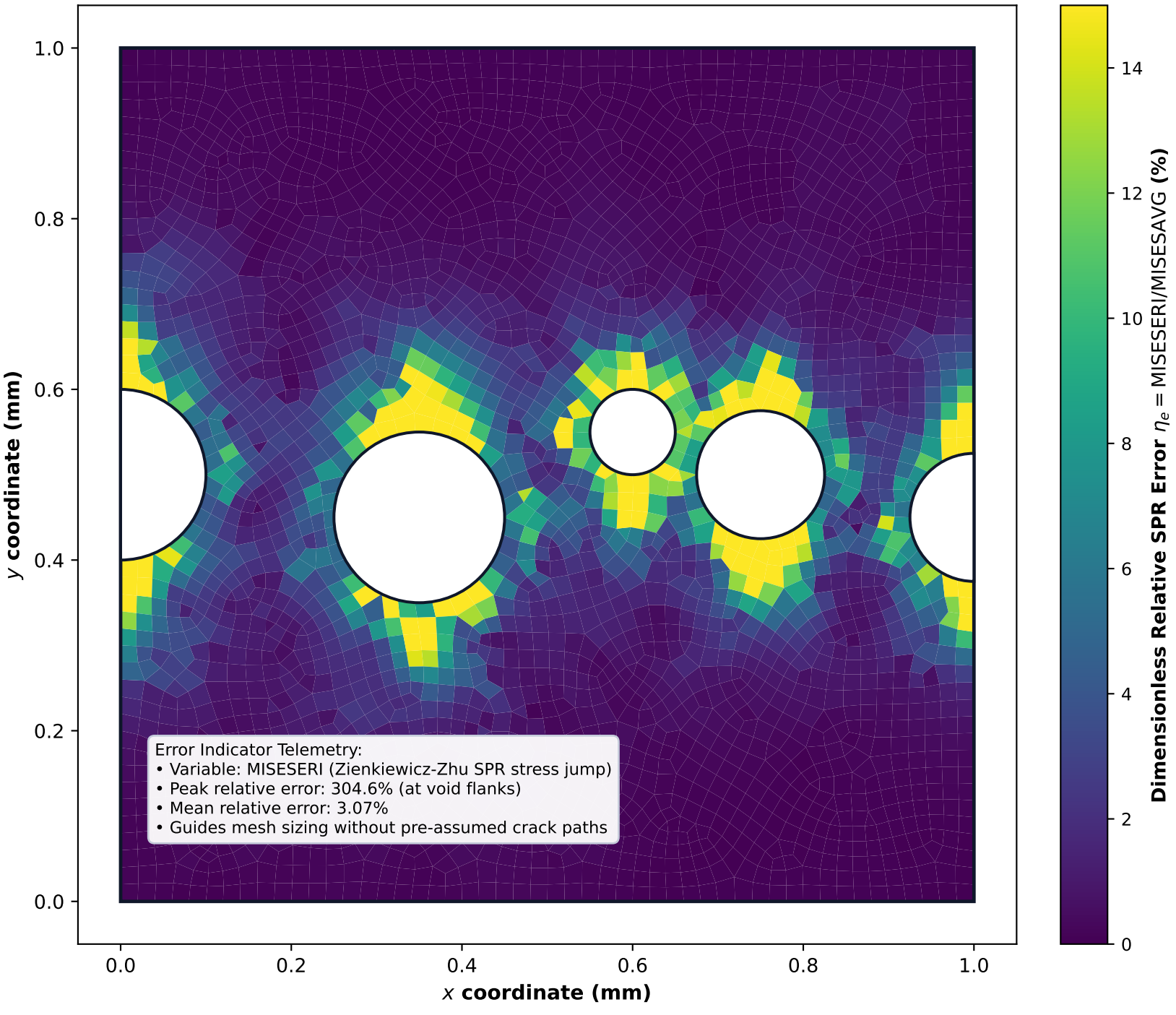

Authentic SPR Stress Jump Error Indicator Field η_e
(Autonomous Localization Along Inter-Hole Ligaments $h_4 \leftrightarrow h_5$, $h_3 \leftrightarrow h_4$, $h_2 \leftrightarrow h_3$)
Error Indicator Telemetry:
• Variable: MISESERI (Zienkiewicz-Zhu SPR stress jump)
• Peak relative error: 304.6% (at void flanks)
• Mean relative error: 3.07%
• Guides mesh sizing without pre-assumed crack paths
y coordinate (mm)
x coordinate (mm)
Dimensionless Relative SPR Error η_e = MISESERI/MISESAVG (%)
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
10
12
14